

GERASIMOS BALATSOUKAS

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Cranfield University

PERSONAL STATEMENT

Motivated and detail-oriented engineer with a background in Biomedical and Mechanical Engineering, combining strong analytical and technical skills with hands-on consulting and research experience. Proven ability to manage data-driven projects, streamline processes, and contribute to multidisciplinary teams. Passionate about applying engineering principles and innovative solutions to address complex technological and organisational challenges. Currently pursuing further studies in order to enhance my theoretical background as well as better familiarise myself with cutting edge engineering processes and software.

EDUCATION

MSc in Advanced Mechanical Engineering, Cranfield University, Cranfield, UK **September 2025 - Current**

- **Modules:** Component Design, Artificial Intelligence for Sustainability, Assessing Risk and Failure, Computational Fluid Dynamics for Renewable Energy, Engineering Design and Project Management, Principals of Engineering, Engineering Stress Analysis: Theory and Simulation, Fluid Mechanics and Loading
- **On Track to complete degree with a 1st Class Honours:** Current average 73%

BEng (Hons) in Biomedical Engineering, University of Glasgow, Glasgow, Scotland **September 2020 - May 2024**

- **Degree Classification:** Upper Second Class with Honours
- **Significant Modules:** Analogue and Digital Electronics, Statics, Dynamics, Fluid Mechanics and Dynamics, Mechanical Design, Mechanics of Structures, Instrumentation and Data Systems, Simulation of Engineering Systems, Control, Biomaterials, Biomechanics, Human Biology, Microscopy and Optics, Medical Imaging, Tissue and Cell Engineering, Biosensors and Diagnostics, Rehabilitation Engineering, Signal Processing of Biosignatures, Introduction to AI
- **Thesis:** Construction of 3-D Piezoelectric Scaffolds for Muscle Regeneration. Final score: 68%
- **Publication:** '3-Dimensional Piezoelectric Fibrous Constructs of Muscle Regeneration.' DOI: <https://doi.org/10.1101/2025.06.19.660455> in BioRxiv and awaiting acceptance after peer review in 'Small' a Nanoscience and Nanotechnology Journal

Highschool Diploma, American Community School Athens, Athens, Greece **September 2017 - May 2020**

- **Results:** International Baccalaureate: 37/45, GPA: 4.388/5, Greek Apolytirion: 17.6/20
- **Classes:** High Level: Mathematics, Physics, Biology and English. Standard Level: Economics and German B
- **Awards:** Community and Service Award

CAREER HISTORY

PwC, Nicosia, Cyprus, Summer Internship in Business Consulting **June 2025 - August 2025**

- Secured two >200,000€ semi-governmental proposals, based on the medical and telecommunications sector, in collaboration with the business consulting team.
- Collaborated on the creation of a map for the merging of two separate HR departments, streamlining the efficiency of the department by 20%, by following best practices found through research.
- Conducted research and analysis to identify benchmarks influenced consulting methodologies, honing skills in developing strategic business proposals and optimising company infrastructure for improved operational capacity.

Fender S.A., Athens, Greek, Data Management of Naval Insurance Contracts **June 2023 - August 2023**

Fender SA is a maritime insurance & risk management company, part of the Cambiaso Risso Group, offering expert coverage for physical, liability, financial and ancillary risks across global naval and marine sectors.

- Streamlined contract data organisation for Fender SA's proprietary system, consolidating Excel sheets, PDFs, and accounting data from international markets (Europe, UK, US, China, and the Middle East) to enhance operational efficiency and asset management, cutting down contract generation times by 50%.
- Corrected mistakes saved in the system of the company, through interpreting multinational insurance contracts and financial data, ensuring parity between contracts and data saved in the system, which significantly reduced mistakes of brokers.

**Agency for Quality Assurance-Greek Ministry of Health, Athens, Greece,
Internship**

June 2022 - August 2022

OAG is a Greek public health authority responsible for setting standards, conducting audits, and implementing initiatives to ensure quality, safety, and resilience in Greece's hospital and healthcare system.

- Conducted technical checks on medical equipment such as defibrillators in hospitals in Thessaloniki and Chania, ensuring adherence to correct guidelines and protocols, as well as assessing the readiness of the equipment for an emergency, under ISO 13485.
- Delivered a report on the best practices for nursing staff training, through statistical analysis on data provided by the Kapodistrian University of Greece, improving training outcomes and KPIs.
- Assigned on the project entitled "Safe Hospitals" with the aim of introduction of directives ensuring SOPs cover all Greek public hospitals from all sorts of emergencies, improving emergency responses by up to 80%, through ensuring adherence to international standards.

BENG ACADEMIC PROJECTS

**University of Glasgow, Scotland, Construction of 3-D Piezoelectric Scaffolds
for muscle Regeneration**

September 2023 - April 2024

- Worked alongside Dr. Oana Dobre in the Advanced Research Centre (ARC) in Glasgow.
- Constructed nanofibres through electrospinning, rolling the fibre complexes with the aim to mimic the geometry of native muscle tissue.
- Assessed the mechano-transudative properties of the fibres, through mechanical, electrical testing and SEM imaging, ensuring the values relevant to the scope of the investigation.
- Treated the fibres, in order to change the polarity, enhancing absorption of laminin and therefore increasing cell adherence.
- Evaluated biocompatibility, cell adherence and cell differentiation through the ZEISS Microscope.
- Published the results on the open-source platform of BioArxiv and submitted the paper on the Small Journal (Awaiting acceptance after completion of peer-review).
- Continued to provide support to three Master level students and on PhD student, in order to further the investigation.

SKILLS, INTERESTS AND EXTRACURRICULAR ACTIVITIES

- **CAD Skills:** Autodesk Fusion 360 (Intermediate), SolidWorks (Intermediate)
- **Simulation Skills:** FEA Abaqus (Intermediate), Ansys Fluent, Design Modeler and ICEM CFD (Intermediate), Simulink (On-Ramp Certified), Ansys Electronics (Intermediate), MOFEM (Beginner), COMSOL Multiphysics (Beginner).
- **Coding Skills:** Python (Advanced), MATLAB (Advanced), HTML 5 & CSS (Intermediate), R (Beginner), Microsoft 365 (Advanced).
- **Further Skills:** 3D/Extrusion Printing (Intermediate), SEM Imaging (Advanced), Cell Culturing (Advanced), Optical Microscopy (Advanced), qPCR for gene expression (Intermediate).
- **Languages:** English: Native, Greek: Native, German: Intermediate with B1 and IB certifications.
- **Extracurricular Activities:** 7 years of kickboxing and free-style wrestling, Motorcycle restorations.